

1. Project Proposal

1.1 Background to the Problem

Managing student-related information is an essential part of university administration. Universities handle a large volume of academic and personal information, including student profiles, faculty information, courses, course enrollment, academic results, schedules, and notices. When these activities are managed manually or through disconnected systems, maintaining accurate, consistent, and easily accessible information becomes increasingly difficult.

Traditional student management processes often require considerable administrative effort and may result in several challenges, including duplicate records, data-entry errors, delays in updating information, difficulty in retrieving student records, and inefficient communication among students, faculty members, and administrators. Students may also experience difficulties in accessing important academic information, such as enrolled courses, academic schedules, examination results, and university notices.

To address these challenges, the proposed **University Student Management System (USMS)** will provide a centralized platform for managing major student-related academic and administrative activities. The system will support three primary user roles: **Administrator, Faculty, and Student**. Each role will be provided with appropriate functionalities and access permissions according to its responsibilities within the university.

The **Administrator** will be responsible for managing student and faculty information, courses, departments, enrollments, academic results, and university notices. **Students** will be able to manage their profiles, view available and enrolled courses, check academic schedules, access examination results and GPA/CGPA information, and view university notices. **Faculty members** will be able to access their assigned courses and students and manage relevant academic information, including student results.

The proposed system aims to reduce manual administrative work, improve data organization and consistency, provide faster access to academic information, and ensure appropriate role-based access to sensitive information. Overall, the University Student Management System is intended to make student-related academic and administrative management more efficient, organized, accessible, and user-friendly.

1.2 Selection of Process Model

For the development of the **University Student Management System**, the **Agile Software Development Model** has been selected as the primary software development process model.

Agile is an iterative and incremental software development approach in which a system is developed through a series of smaller development cycles known as iterations. Instead of developing the complete system in a single sequence, the project is divided into manageable modules or features. Each iteration involves activities such as planning, development, testing, review, and improvement before proceeding to subsequent iterations.

The Agile model is suitable for this project because the University Student Management System consists of several independent but interconnected modules, including user authentication, student management, faculty management, course management, course enrollment, attendance, result management, and notice management. These modules can be developed incrementally and integrated throughout the development process.

Another important reason for selecting Agile is its flexibility in handling changing requirements. During the development of an academic project, requirements or implementation decisions may change based on feedback from the instructor, project team members, or testing results. The Agile model allows such changes to be incorporated into subsequent iterations without requiring the entire system to be redesigned from the beginning.

The proposed development process can be organized into several iterations. The initial iteration may focus on system authentication and user roles. Subsequent iterations may cover student and faculty management, course and enrollment management, attendance and result management, notice management, and finally system integration, testing, and refinement. This approach allows the development team to gradually build and validate the complete system.

The major reasons for selecting the Agile Software Development Model are as follows:

- **Incremental Development:** The system can be developed and delivered module by module, allowing the team to focus on manageable components.
- **Flexibility:** Changes or improvements in requirements can be incorporated into subsequent iterations according to project needs and feedback.
- **Continuous Testing:** Each module can be tested during its development cycle, allowing errors and inconsistencies to be identified at an early stage.
- **Early Feedback:** Feedback from instructors, team members, and testing activities can be incorporated throughout the development process.
- **Risk Reduction:** Potential technical and functional problems can be identified and addressed earlier rather than being discovered only at the final stage.
- **Team Collaboration:** Team members can work collaboratively on different modules while maintaining an organized development workflow.
- **Continuous Improvement:** Existing features can be reviewed, refined, and improved throughout the development lifecycle.

Therefore, the **Agile Software Development Model** is considered an appropriate process model for the University Student Management System because it supports incremental development, continuous testing, flexibility, early feedback, effective collaboration, and continuous improvement. These characteristics are well suited to the development of a modular university management system within an academic project environment.

2. Timeline Chart / Gantt Chart

The Timeline Chart, also known as the Gantt Chart, represents the planned schedule of the **University Student Management System (USMS)** project. It provides a visual representation of the activities that need to be completed during the project development period and shows the expected duration of each activity.

The project has been planned over a period of **13 weeks**. Each activity is assigned to a specific team member, and the corresponding weeks indicate when that particular activity will be performed. The Gantt Chart helps the team maintain a structured workflow, monitor project progress, identify overlapping activities, and ensure that all major project tasks are completed within the planned timeframe.

The project team consists of **Abir, Othoi, Mahir, Tanjim, and Hadiul**. Responsibilities are distributed among the team members according to the requirements and development phases of the project.

2.1 Project Timeline

The major activities of the University Student Management System are planned as follows:

Task - Person	Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13
A : Abir														
B : Othoi														
C : Mahir														
D : Tanjim														
E : Tanjim														
F : Abir														
G : Hadiul														
H : Othoi														
I : Hadiul														
J : Mahir														
K : Abir														

Activity key:

A : Project Proposal
B : Requirement Analysis
C : SRS / PRD Preparation
D : User Stories & RTM
Project members: Abir, Othoi, Mahir, Tanjim, Hadiul

E : System Design
F : UI Design using Figma
G : System Development
H : Git Workflow

I : Integration & Testing
J : Bug Fixing & Refinement
K : Documentation & Final Submission



2.2 Detailed Weekly Schedule

Week 1 – Project Proposal and Initial Requirement Analysis

During the first week, the project will be formally initiated. Abir will prepare the initial project proposal for the University Student Management System. The proposal will include the project background, problem statement, objectives, project scope, expected outcomes, and an overview of the proposed system.

At the same time, Othoi will begin the initial requirement analysis. The purpose of this phase is to understand the problems related to traditional university student management and identify the requirements of different users such as students, teachers, and administrators.

Week 2 – Requirement Analysis and SRS / PRD Preparation

During Week 2, requirement gathering and analysis will continue. Othoi will identify and organize the major functional requirements of the system. These may include student registration, student profile management, course management, teacher information, course enrollment, attendance management, result management, and notice management.

Mahir will begin preparing the Software Requirements Specification (SRS) / Product Requirements Document (PRD) based on the requirements identified during the analysis phase.

Week 3 – SRS / PRD and User Stories

During Week 3, the requirements will be refined and documented in a structured manner. Mahir will continue working on the SRS / PRD and ensure that the major system requirements are properly documented.

Tanjim will begin preparing the User Story Table and Requirements Traceability Matrix (RTM). User stories will describe the system requirements from the perspective of different users.

For example:

“As a student, I want to view my enrolled courses so that I can easily manage my academic information.”

Similar user stories will be prepared for students, teachers, and administrators.

Week 4 – User Stories, System Design and Database Design

During Week 4, Tanjim will complete the major User Stories and Requirements Traceability Matrix. After completing the requirements-related activities, the project will move toward the system design phase.

Tanjim will work on the overall System Design, including the system architecture, major modules, user roles, data flow, and relationships between different components of the system.

At the same time, Abir will begin the Database Design. The database structure will be planned based on the requirements and system modules.

Week 5 – Database Design, UI Design, System Development and Git Workflow

During Week 5, multiple project activities will take place simultaneously.

Abir will continue working on the Database Design and UI Design using Figma. The database design will identify important entities such as Student, Teacher, Course, Department, Enrollment, Attendance, Result, Notice, and User/Admin.

The relationships between these entities will also be defined to ensure that the database can properly support the system requirements.

Abir will also work on the UI design in Figma. The major screens of the system may include Login, Student Dashboard, Student Profile, Course Management, Attendance, Result, Notice, and Admin Dashboard.

Hadiul will begin the main system development based on the approved requirements and system design.

Othoi will begin managing the Git workflow and version-control process to ensure that the team's source code and development progress are properly maintained.

Week 6 – UI Finalization and System Development

During Week 6, Abir will finalize the major UI designs in Figma. The design will focus on creating a consistent, simple, and user-friendly interface for the University Student Management System.

Hadiul will continue system development based on the finalized system design and UI structure. The development process will focus on implementing the major functional modules of the system.

Week 7 – System Development

During Week 7, the primary focus will be system development. Hadiul will continue implementing the core features of the University Student Management System.

The development may include modules such as authentication, student management, course management, teacher management, enrollment management, attendance management, result management, and notice management.

Othoi will continue maintaining the Git workflow and ensuring that development changes are properly tracked.

Week 8 – System Development

During Week 8, system development will continue. The major modules will be connected and refined so that they can work together as a complete system.

The development team will also verify that the implemented features follow the requirements defined in the SRS / PRD.

Week 9 – System Development and Integration

During Week 9, the main development phase will move toward completion. Hadiul will finalize the remaining major development tasks and begin integrating the different system modules.

Integration will ensure that individual modules can communicate and operate together properly.

For example, a student should be able to log in, access the dashboard, view courses, check enrollment information, and view academic results through a connected system.

Week 10 – Integration, Testing and Bug Fixing

During Week 10, the project will focus heavily on integration and testing. Hadiul will continue Integration and Testing to verify whether the different modules work together correctly.

Mahir will begin identifying and fixing bugs found during testing.

Testing may include functional testing, integration testing, UI testing, input validation, error handling, and basic usability testing.

Week 11 – Testing, Bug Fixing and Documentation

During Week 11, integration testing will continue to ensure that the system performs according to the specified requirements.

Mahir will continue bug fixing and refinement. Any errors, unexpected behavior, or usability problems discovered during testing will be corrected.

At the same time, Abir will begin preparing the final project documentation. The documentation may include the system overview, requirements, system design, database design, UI screenshots, testing results, Git workflow, limitations, and future improvements.

Week 12 – Bug Fixing and Documentation

During Week 12, the project team will focus on final refinement and documentation.

Mahir will complete the remaining bug fixes and make necessary improvements to the system.

Abir will continue finalizing the project documentation and ensure that all required sections of the project report are properly organized.

The team will also verify that the final system is consistent with the requirements defined in the SRS / PRD.

Week 13 – Final Submission and Presentation

Week 13 will be dedicated to the final preparation and submission of the project.

During this week, the team will perform the final system check, verify the major functionalities, complete the documentation, prepare presentation materials, organize project files, verify the GitHub repository, and prepare the final project demonstration.

Abir will coordinate the final documentation and submission process with the other team members.

2.3 Purpose of the Gantt Chart

The Gantt Chart is an important project management tool for the University Student Management System because it provides a clear overview of what needs to be done, who is responsible for each activity, and when the activity is expected to be completed.

The major purposes of the Gantt Chart are described below:

1. **Task Planning:**

The Gantt Chart divides the entire project into smaller and manageable activities. This makes the overall development process easier to understand and organize.

2. **Time Management:**

Each activity is assigned a specific time period. This allows the team to manage the available 13 weeks efficiently.

3. **Responsibility Assignment:**

Each major task is assigned to a specific team member. This ensures that every member understands their responsibilities.

4. **Progress Monitoring:**
The team can compare the planned schedule with the actual progress of the project. This helps identify whether any activity is delayed.
5. **Parallel Activities:**
Some activities can be performed simultaneously. For example, database design, UI design, development, and Git workflow can overlap without unnecessarily delaying the project.
6. **Risk Reduction:**
Proper scheduling helps the team identify possible delays and provides sufficient time for testing and bug fixing before the final submission.
7. **Better Team Coordination:**
Since every team member has assigned responsibilities and deadlines, the Gantt Chart helps improve communication and coordination among the team members.
8. **Final Submission Planning:**
Documentation, final testing, and presentation preparation are scheduled before the final submission so that the team has sufficient time to make necessary corrections.

2.4 Overall Project Development Flow

The overall development process of the University Student Management System follows a systematic sequence:

Project Proposal → Requirement Analysis → SRS / PRD → User Stories & RTM → System Design → Database Design & UI Design → System Development → Integration & Testing → Bug Fixing & Refinement → Documentation → Final Submission & Presentation

This planned timeline ensures that the University Student Management System is developed in a structured and systematic manner. The project starts with proper planning and requirement analysis, followed by documentation, system and database design, UI design, implementation, integration, testing, bug fixing, and finally documentation and submission.

The Gantt Chart therefore provides a clear roadmap for the entire project and helps the team complete the University Student Management System within the planned 13-week schedule.

3. Software Requirements Specification (SRS)

3.1 Scopes and Features

3.1.1 Scope

The **University Student Management System (USMS)** is a centralized software system designed to manage essential student and academic information in an organized, efficient, and accessible manner. The system aims to reduce dependency on manual record-keeping, improve information accessibility, and maintain academic records through a structured digital platform.

The system will support three primary user roles: **Administrator, Faculty, and Student**. Each role will have specific permissions and functionalities based on its responsibilities within the university.

The scope of the system includes:

- Student information management
- Faculty information management
- Department and course management
- Course enrollment management
- Attendance management
- Academic result and grade management
- Academic schedule management
- Notice and announcement management
- User authentication and role-based access control
- Role-specific dashboards
- Profile management
- Search and information retrieval
- Centralized database management
- Data validation and error handling

The initial implementation will focus on the core academic and administrative requirements of student management. Advanced services such as online payment, biometric attendance, facial recognition, dedicated mobile applications, and AI-based academic analytics are excluded from the initial scope.

3.1.2 Major Features

1. User Authentication and Authorization

The system shall provide secure authentication for registered users and enforce role-based access control. Students, faculty members, and administrators shall have access only to the functions permitted for their respective roles.

2. Student Management

The system shall maintain student records including student ID, name, contact information, department, academic program, semester, and academic status. Authorized administrators shall be able to create, view, update, and manage student records.

3. Faculty Management

The system shall maintain faculty information such as faculty ID, name, department, designation, contact information, and assigned courses. Administrators shall be responsible for maintaining faculty records.

4. Department and Course Management

The system shall maintain department and course information in a structured manner. Course records shall include relevant details such as course code, course name, credit hours, department, semester, and assigned faculty member.

5. Course Enrollment Management

The system shall maintain student enrollment records and associate students with their respective courses and academic semesters.

6. Attendance Management

Faculty members shall be able to record attendance for their assigned courses, while students shall be able to view their attendance records and attendance status.

7. Result and Grade Management

Authorized faculty members or administrators shall be able to manage student marks, grades, and grade points. Students shall be able to view their academic results and GPA/CGPA information.

8. Academic Schedule Management

The system shall provide students and faculty members with access to relevant academic schedules, including course schedules, class information, and related academic activities.

9. Notice Management

Administrators or authorized users shall be able to publish academic and administrative notices. Students and faculty members shall be able to view relevant notices through the system.

10. Role-Based Dashboard

The system shall provide separate dashboards for students, faculty members, and administrators. Each dashboard shall display information and functions relevant to the user's role.

11. Profile Management

Users shall be able to view their profile information and update permitted personal details. Restricted academic and administrative information shall remain accessible only to authorized users.

12. Search and Information Retrieval

The system shall provide search functionality to enable authorized users to efficiently retrieve relevant student, faculty, course, and academic records.

13. Centralized Database Management

The system shall maintain a centralized database containing user, student, faculty, department, course, enrollment, attendance, result, schedule, and notice information. Appropriate relationships shall be maintained to ensure data consistency and integrity.

14. Data Validation and Error Handling

The system shall validate user input before processing or storing information. Appropriate error messages shall be provided for invalid, incomplete, or unauthorized operations.

3.1.3 User-Specific Scope

User Role	Primary Responsibilities
Student	View and manage permitted profile information, enrolled courses, attendance, results, GPA/CGPA, academic schedules, and university notices.
Faculty	View assigned courses and students, manage attendance and academic results, and access relevant academic information and notices.
Administrator	Manage students, faculty, departments, courses, enrollments, users, notices, schedules, and academic records.

3.1.4 In-Scope

The initial version of the University Student Management System will include:

1. User authentication and authorization
2. Role-based access control
3. Student information management
4. Faculty information management
5. Department management
6. Course management
7. Course enrollment management
8. Attendance management
9. Result and grade management
10. Academic schedule management
11. Notice management
12. Role-based dashboards
13. Profile management
14. Search and information retrieval
15. Centralized database management
16. Data validation and error handling

17. System testing and validation

3.1.5 Out-of-Scope

The following functionalities are excluded from the initial implementation:

- Online tuition fee payment
- Online banking or payment gateway integration
- Biometric or fingerprint-based attendance
- Facial recognition
- Dedicated Android or iOS mobile application
- SMS gateway integration
- AI-based academic performance prediction
- Advanced business intelligence and predictive analytics
- Integration with external university ERP systems

These functionalities may be considered as **future enhancements** if the system is further developed.

3.1.6 Expected Benefits

The proposed system is expected to provide the following benefits:

- Reduction of manual paperwork and administrative workload
- Improved accuracy and consistency of academic records
- Faster access to student and academic information
- Simplified attendance and result management
- Improved communication through centralized notices
- Controlled access to sensitive academic information
- Better organization of university student records
- Improved efficiency of academic and administrative processes

Overall, the University Student Management System will provide a **centralized, secure, and role-based platform** for managing the core academic and administrative information associated with university students.

3.2 User Story Table

User stories describe the functional expectations of the **University Student Management System** from the perspective of its intended users. They identify what each user wants to accomplish through the system and the purpose of performing the corresponding action.

The system has three primary user roles: **Student, Faculty, and Administrator**. Each role has different responsibilities and access privileges. The user stories are therefore organized according to these three user categories.

Each user story follows the standard format:

As a [user role], I want to [perform an action], so that [achieve a specific objective].

The identified user stories will serve as a foundation for defining functional requirements, system design, implementation, and testing.

3.2.1 Student User Stories

Students will primarily use the system to access and monitor their personal and academic information.

ID	User Story	Priority
US-01	As a student, I want to log in securely so that I can access my authorized academic information.	High
US-02	As a student, I want to view my profile so that I can verify my personal and academic information.	High
US-03	As a student, I want to update permitted personal information so that my profile remains accurate and up to date.	Medium
US-04	As a student, I want to view my enrolled courses so that I can keep track of my current academic courses.	High
US-05	As a student, I want to view course details so that I can understand the information related to my courses.	Medium
US-06	As a student, I want to view my attendance records so that I can monitor my attendance status.	High
US-07	As a student, I want to view my academic results so that I can monitor my performance in different courses.	High
US-08	As a student, I want to view my GPA/CGPA so that I can evaluate my overall academic performance.	Medium
US-09	As a student, I want to view my academic schedule so that I can keep track of classes and related activities.	High

ID	User Story	Priority
US-1 0	As a student, I want to view university notices so that I can remain informed about important announcements.	High

3.2.2 Faculty User Stories

Faculty members will use the system to perform academic activities related to their assigned courses and students.

ID	User Story	Priority
US-1 1	As a faculty member, I want to log in securely so that I can access the functions associated with my role.	High
US-1 2	As a faculty member, I want to view my assigned courses so that I can manage the courses under my responsibility.	High
US-1 3	As a faculty member, I want to view students enrolled in my courses so that I can manage their academic activities.	High
US-1 4	As a faculty member, I want to record student attendance so that attendance information can be maintained accurately.	High
US-1 5	As a faculty member, I want to update student results so that academic performance can be recorded properly.	High
US-1 6	As a faculty member, I want to view my academic schedule so that I can manage my assigned classes and activities.	Medium
US-1 7	As a faculty member, I want to view relevant university notices so that I can remain informed about academic and administrative activities.	Medium

3.2.3 Administrator User Stories

Administrators will have the broadest access to the system and will be responsible for maintaining major academic and administrative records.

ID	User Story	Priority
US-1 8	As an administrator, I want to log in securely so that I can access administrative functions.	High
US-1 9	As an administrator, I want to add and manage student records so that student information remains organized and accurate.	High
US-2 0	As an administrator, I want to add and manage faculty records so that faculty information remains accurate and up to date.	High

ID	User Story	Priority
US-2 1	As an administrator, I want to manage departments so that academic information can be properly organized.	Medium
US-2 2	As an administrator, I want to create and manage courses so that course information remains accurate and current.	High
US-2 3	As an administrator, I want to manage student enrollment records so that course registration information remains organized.	High
US-2 4	As an administrator, I want to manage user roles and permissions so that system access can be properly controlled.	High
US-2 5	As an administrator, I want to publish and manage notices so that important information can be communicated to students and faculty members.	High
US-2 6	As an administrator, I want to manage academic schedules so that course and class information remains organized.	Medium
US-2 7	As an administrator, I want to search and retrieve system records so that required information can be accessed efficiently.	Medium
US-2 8	As an administrator, I want to manage academic records so that student information remains accurate and consistent.	High

3.2.4 Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered successfully implemented.

US-01: Student Login

User Story:

As a student, I want to log in securely so that I can access my authorized academic information.

Acceptance Criteria:

- The student must provide valid login credentials.
- The system must authenticate the provided credentials.
- Valid credentials must provide access to the student dashboard.
- Invalid credentials must generate an appropriate error message.
- Students must not be able to access unauthorized faculty or administrator functions.

US-06: View Attendance

User Story:

As a student, I want to view my attendance records so that I can monitor my attendance status.

Acceptance Criteria:

- Only authenticated students can access attendance information.
- The system must display attendance for the student's enrolled courses.
- Attendance information must correspond to the correct student and course.
- Students must not be permitted to modify attendance records.

US-14: Record Attendance

User Story:

As a faculty member, I want to record student attendance so that attendance information can be maintained accurately.

Acceptance Criteria:

- Only authorized faculty members can access attendance management.
- The faculty member must be able to select an assigned course.
- The system must display students enrolled in the selected course.
- The faculty member must be able to record attendance.
- Attendance records must be stored correctly.
- Faculty members must not be permitted to modify attendance for unauthorized courses.

US-15: Update Results

User Story:

As a faculty member, I want to update student results so that academic performance can be recorded properly.

Acceptance Criteria:

- Only authorized faculty members can access result management.
- The faculty member must be able to select an assigned course.
- Result information must be associated with the correct student and course.
- Marks and grades must be validated before being stored.
- Students must be able to view updated results.
- Unauthorized users must not be able to modify results.

US-19: Student Management

User Story:

As an administrator, I want to add and manage student records so that student information remains organized and accurate.

Acceptance Criteria:

- Only authorized administrators can manage student records.
- The administrator must be able to add a new student record.

- Required student information must be validated.
- Student IDs must be unique.
- The administrator must be able to update existing records.
- Unauthorized users must not be permitted to modify administrative student records.

US-22: Course Management

User Story:

As an administrator, I want to create and manage courses so that course information remains accurate and current.

Acceptance Criteria:

- Only authorized administrators can manage course information.
- Required course information must be provided.
- Course codes must be unique.
- The administrator must be able to create and update course records.
- Course information must be stored correctly in the database.

3.2.5 User Story Prioritization

The user stories are prioritized according to their importance to the core operation of the system.

High Priority

High-priority user stories represent essential functionalities required for the basic operation of the University Student Management System. These include:

- User authentication
- Role-based authorization
- Student management
- Faculty management
- Course management
- Course enrollment
- Attendance management
- Result management
- Notice management
- User and permission management

Medium Priority

Medium-priority user stories support usability and additional system functionality. These include:

- Profile updates
- Course detail viewing
- GPA/CGPA viewing
- Academic schedule viewing
- Search and information retrieval
- System monitoring and related administrative functions

3.2.6 Summary

The User Story Table provides a structured representation of the requirements of the three primary user groups of the University Student Management System. It establishes a clear relationship between **user roles, required system interactions, priorities, and expected outcomes**.

These user stories will be used as a basis for developing the system's functional requirements and Requirements Traceability Matrix. They will also support system design, implementation, and acceptance testing.

3.3 Requirements Traceability Matrix (RTM)

The **Requirements Traceability Matrix (RTM)** establishes a systematic relationship between user stories, functional requirements, and corresponding system features. It ensures that each identified requirement is properly addressed throughout the design, development, and testing phases of the University Student Management System.

The RTM also helps the development team identify missing requirements, maintain requirement consistency, and verify that the implemented system satisfies the identified user needs.

3.3.1 Functional Requirements Traceability Matrix

Requirement ID	User Story	Requirement Description	Priority	Related Feature
FR-01	US-01, US-11, US-18	The system shall authenticate users using valid login credentials.	High	User Authentication
FR-02	US-01, US-11, US-18	The system shall provide role-based access according to the user's assigned role.	High	Authorization
FR-03	US-02, US-19	The system shall maintain and display student profile information.	High	Student Management

Requirement ID	User Story	Requirement Description	Priority	Related Feature
FR-04	US-03	The system shall allow students to update permitted personal information.	Medium	Profile Management
FR-05	US-19	The system shall allow administrators to create, view, update, and manage student records.	High	Student Management
FR-06	US-20	The system shall allow administrators to create, view, update, and manage faculty records.	High	Faculty Management
FR-07	US-21	The system shall allow administrators to create and manage department information.	Medium	Department Management
FR-08	US-22	The system shall allow administrators to create, update, and manage course information.	High	Course Management
FR-09	US-04, US-23	The system shall maintain student course enrollment records.	High	Course Enrollment
FR-10	US-05, US-12	The system shall display relevant course information to authorized users.	Medium	Course Management
FR-11	US-06	The system shall allow students to view their attendance records.	High	Attendance Management
FR-12	US-14	The system shall allow faculty members to record and update attendance for assigned courses.	High	Attendance Management
FR-13	US-07	The system shall allow students to view their academic results.	High	Result Management
FR-14	US-08	The system shall display GPA/CGPA information to authorized students.	Medium	Result Management
FR-15	US-15	The system shall allow authorized faculty members to enter and update student results.	High	Result Management
FR-16	US-10, US-17, US-25	The system shall allow authorized users to publish and view university notices.	High	Notice Management
FR-17	US-09, US-16, US-26	The system shall provide authorized users with access to relevant academic schedules.	Medium	Schedule Management

Requirement ID	User Story	Requirement Description	Priority	Related Feature
FR-18	US-27	The system shall provide search functionality for authorized records.	Medium	Search and Retrieval
FR-19	US-24	The system shall allow administrators to manage user roles and permissions.	High	User Management
FR-20	US-28	The system shall maintain academic records in a centralized database.	High	Database Management
FR-21	US-01–US-28	The system shall validate user input before processing or storing information.	High	Data Validation
FR-22	US-01–US-28	The system shall restrict unauthorized access to protected information and functions.	High	Security
FR-23	US-18, US-24	The system shall provide administrators with access to relevant system information for management purposes.	Medium	Administration

3.3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics that the University Student Management System must satisfy. These requirements describe **how the system should perform** rather than specifying individual system functions.

Requirement ID	Category	Non-Functional Requirement	Priority
NFR-01	Security	The system shall protect user and academic information from unauthorized access.	High
NFR-02	Performance	The system should provide an acceptable response time under the expected project workload.	High
NFR-03	Usability	The system interface should be clear, consistent, and easy for users to navigate.	High
NFR-04	Reliability	The system should perform core operations consistently without unexpected failures.	High
NFR-05	Availability	The system should remain available during normal operating periods.	Medium

Requirement ID	Category	Non-Functional Requirement	Priority
NFR-06	Data Integrity	The system shall maintain accurate, consistent, and reliable data across related records.	High
NFR-07	Maintainability	The system should be structured to support future maintenance and modification.	Medium
NFR-08	Scalability	The system should be capable of accommodating an increasing number of users and records with appropriate system improvements.	Medium
NFR-09	Compatibility	The system should operate correctly on commonly used modern web browsers and supported devices.	Medium
NFR-10	Privacy	Personal and academic information shall be accessible only to authorized users.	High

3.3.3 Traceability Relationship

The relationship between user requirements and system implementation can be represented as follows:

User Story → Functional Requirement → System Feature → Implementation → Testing

For example:

US-14:

As a faculty member, I want to record student attendance so that attendance information can be maintained accurately.

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FR-12:

The system shall allow faculty members to record and update attendance for assigned courses.

↓

System Feature:

Attendance Management

↓

Implementation:

Attendance Management Module

↓

Testing:

Attendance-related test cases

This traceability relationship ensures that each major user requirement is translated into a specific system requirement and can subsequently be verified through appropriate testing activities.

3.3.4 Requirement Prioritization

The requirements are categorized according to their importance to the successful operation of the system.

High-Priority Requirements

High-priority requirements are essential to the core functionality of the University Student Management System. These include:

- User authentication and authorization
- Student management
- Faculty management
- Course management
- Course enrollment
- Attendance management
- Result management
- Notice management
- User and permission management
- Data validation
- Security
- Data integrity

Medium-Priority Requirements

Medium-priority requirements support usability, accessibility, and additional administrative functionality. These include:

- Profile management
- Course information viewing
- GPA/CGPA display
- Academic schedule viewing
- Search functionality
- System monitoring
- Maintainability

- Scalability
- Compatibility

3.3.5 Purpose of the RTM

The Requirements Traceability Matrix will be used throughout the project to:

- Ensure that all identified requirements are addressed.
- Establish a clear relationship between user needs and system functionality.
- Support systematic system design and development.
- Prevent important requirements from being overlooked.
- Provide a basis for developing test cases.
- Verify that implemented features satisfy their corresponding requirements.
- Support project monitoring and documentation.
- Maintain consistency between requirements, implementation, and testing.

3.3.6 Summary

The Requirements Traceability Matrix provides a structured mechanism for linking the **user stories, functional requirements, non-functional requirements, and system features** of the University Student Management System.

Through traceability, each major requirement can be followed from its original user need through implementation and testing. This helps ensure **requirement completeness, consistency, verification, and effective quality control** throughout the software development lifecycle.

4...UI/figma

5.github

Conclusion

The proposed **University Student Management System (USMS)** provides a structured and centralized approach to managing essential student-related academic and administrative activities. The system is designed to address the limitations of manual and fragmented

management processes by integrating student, faculty, department, course, enrollment, attendance, result, schedule, and notice management into a unified platform.

The defined scope, user stories, functional requirements, and non-functional requirements establish a clear foundation for the development of the proposed system. The identification of role-based responsibilities for **Administrators, Faculty Members, and Students** ensures that system functionalities and access privileges are aligned with the specific requirements of each user group. Furthermore, the Requirements Traceability Matrix provides a systematic relationship between user needs, system requirements, features, implementation, and testing.

The selection of the **Agile Software Development Model** will support the incremental development, continuous testing, feedback incorporation, and refinement of the system throughout the project lifecycle. This approach will also enable the development team to manage the different modules effectively while maintaining flexibility in response to changes or improvements in requirements.

Overall, the proposed University Student Management System is expected to improve the efficiency, accuracy, accessibility, and organization of student-related academic management. By providing a centralized and role-based digital platform, the system can reduce manual effort, improve information management, and facilitate more effective interaction among students, faculty members, and administrators. The proposed requirements and project structure therefore provide a suitable foundation for the subsequent phases of **system design, database development, implementation, testing, and evaluation**.